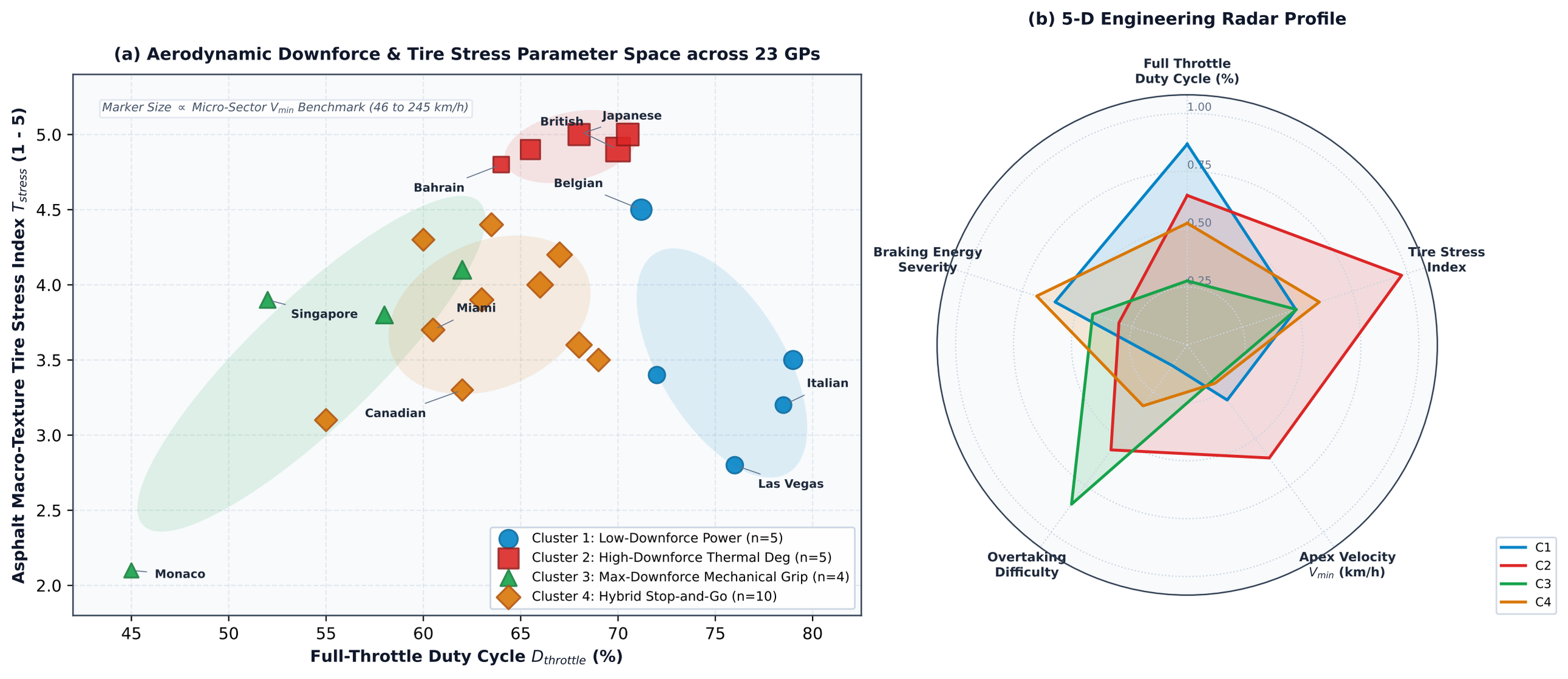

(a) Aerodynamic Downforce & Tire Stress Parameter Space across 23 GPs
Marker Size ∝ Micro-Sector V_{min} Benchmark (46 to 245 km/h)
British
Japanese
Bahrain
Belgian
Singapore
Miami
Canadian
Italian
Las Vegas
Monaco
Cluster 1: Low-Downforce Power (n=5)
Cluster 2: High-Downforce Thermal Deg (n=5)
Cluster 3: Max-Downforce Mechanical Grip (n=4)
Cluster 4: Hybrid Stop-and-Go (n=10)
Asphalt Macro-Texture Tire Stress Index T_{stress} (1 - 5)
Full-Throttle Duty Cycle $D_{throttle}$ (%)
2.0
2.5
3.0
3.5
4.0
4.5
5.0
45
50
55
60
65
70
75
80
(b) 5-D Engineering Radar Profile
Full Throttle Duty Cycle (%)
Tire Stress Index
Apex Velocity V_{min} (km/h)
Overtaking Difficulty
Braking Energy Severity
1.00
0.75
0.50
0.25
C1
C2
C3
C4